

# Multi-Agent Research System

A powerful multi-agent research pipeline built with LangChain that automates the research process through specialized agents. The system takes a research topic and generates a comprehensive, structured report with quality evaluation.

## Features

- **Web Search Agent** - Uses Tavily API to find recent, relevant information
- **Content Reader Agent** - Scrapes and extracts detailed content from URLs using BeautifulSoup
- **Report Writer Chain** - Generates professional, structured research reports
- **Critic Chain** - Evaluates reports with scores and actionable feedback
- **Dual Interfaces** - CLI and Streamlit web UI

## Architecture

User Input (topic)

Search Agent  
(Tavily API)

Reader Agent  
(BeautifulSoup)

Writer Chain (LCEL)  
Report Generation

Critic Chain (LCEL)  
Quality Evaluation

Final Report + Feedback

## Installation

```
# Clone the repository
cd multi_agent_system

# Create virtual environment
python -m venv .venv
.venv\Scripts\activate # Windows
# source .venv/bin/activate # Linux/Mac

# Install dependencies
pip install -r requirements.txt
```

## Configuration

Create a `.env` file in the project root with your API keys:

```
TAVILY_API_KEY=your_tavily_api_key
OPENROUTER_API_KEY=your_openrouter_api_key
# Optional: OPENAI_API_KEY=your_openai_key
```

Get your API keys: - Tavily - Search API - OpenRouter - LLM API (free tier available)

## Usage

### Command Line

```
python pipeline.py
```

### Streamlit Web UI

```
streamlit run app.py
```

## Project Structure

| File             | Description                               |
|------------------|-------------------------------------------|
| tools.py         | Tool definitions (web_search, scrape_url) |
| agents.py        | Agent and LCEL chain definitions          |
| pipeline.py      | Core research pipeline orchestration      |
| app.py           | Streamlit web interface                   |
| main.py          | Entry point                               |
| requirements.txt | Python dependencies                       |

## Tech Stack

- **LangChain** - Agent framework and LCEL chains

- **OpenRouter** - LLM API (gpt-oss-20b:free model)
- **Tavily** - Web search
- **BeautifulSoup** - Web scraping
- **Streamlit** - Web UI
- **Python 3.13+**

## **License**

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